

Item Bank — ER Diagram & UML Class Model (Phase 1)

Technical Architecture Specification • Phase 1 Item Bank • Relational Schema, Multiplicities & Security Perimeter

NOTATION LEGEND: [PK] Primary Key [FK] Foreign Key (1:1) One-to-One (1:N) One-to-Many (M:N) Many-to-Many [LOCKED] Hidden from Candidate API (Zero-Knowledge)



Item Bank Relational Schema & Security Visibility Matrix

Comprehensive table definitions, constraints, candidate API exposure boundaries, and psychological purpose.

Table Name	Column Name	Data Type	Constraints	API Visibility	Architectural Purpose & Security Rule
battery	id / code / name_ar	BIGINT/VARCHAR	PK / NOT NULL	Public	Master assessment battery catalog (PQ10, SJT, Derailer, GCAT).
battery	weight_percent	DECIMAL(5,2)	NOT NULL	Admin/Scoring	Composite score weighting (PQ10: 28%, SJT: 22%, Derail: 20%, GCAT: 30%).
competency	id / name_ar / def_ar	BIGINT/VARCHAR/TEXT	PK / NOT NULL	Report/Admin	10 Core Leadership competencies (e.g. Strategic Thinking, Execution).
cognitive_domain	type / topic_ar	ENUM/VARCHAR	NOT NULL	Candidate	GCAT Cognitive Aptitude domains (VERBAL, NUMERICAL, ABSTRACT).
difficulty_level	code	VARCHAR(50)	UNIQUE, NOT NULL	Admin	Adaptive item bank tiering (EASY, MEDIUM, HARD).
personality_item	statement_ar	TEXT	NOT NULL	Candidate	140 PQ10 Likert statements presented to candidates during assessment.
personality_item	is_reverse_keyed	BOOLEAN	NOT NULL	[LOCKED] HIDDEN	Scoring direction flag. Inverts response scale (1->5, 5->1).
personality_item	ideal_target	INT	NOT NULL	[LOCKED] HIDDEN	Psychometric benchmark target for norm calibration and deviation calculation.
derailer_item	statement_ar / type	TEXT/VARCHAR	NOT NULL	Candidate	60 Dark Triad statements (Volatility, Micromanagement, Arrogance).
derailer_item	ideal_target	INT	NOT NULL	[LOCKED] HIDDEN	Derailment risk baseline. Scores deviating beyond target flag risk bands.
cognitive_item	question_text_ar	TEXT	NOT NULL	Candidate	42 GCAT aptitude problem statements (matrices, syllogisms, series).
cognitive_item	correct_option_letter	CHAR(1)	NOT NULL	[LOCKED] HIDDEN	Correct key ('A'..'D'). Must NEVER be transmitted to candidate client.
cognitive_item_option	option_letter / text	CHAR/TEXT	FK, NOT NULL	Candidate	4 multiple-choice alternatives per cognitive problem.
sjt_scenario	situation_text_ar	TEXT	NOT NULL	Candidate	16 Realistic workplace leadership dilemmas and critical signals.
sjt_option	expert_rating	DECIMAL(3,2)	NOT NULL	[LOCKED] HIDDEN	SME benchmark consensus weight (1.00..4.00) for Kendall-tau correlation.
sjt_scenario_analysis	best_option_letter	CHAR(1)	NOT NULL	[LOCKED] HIDDEN	Gold-standard expert resolution letter and psychometric rationale.
battery_selection_rule	grouping / items_req	ENUM/INT	FK, NOT NULL	Engine Internal	Item assembly constraints (e.g. 14 items per competency for PQ10).
test_session_item	item_type / item_id	ENUM/BIGINT	FK, INDEXED	Candidate/Engine	Hydrated test session item queue with randomized sequence order.
item_usage_log	item_type / used_at	ENUM/TIMESTAMP	INDEXED	Admin/Analytics	Item exposure and fatigue audit log.

Item Bank UML Class Model (Spring Boot / JPA Entity Layer)

Object-oriented domain model with JPA relationship mappings, typed attributes, and discriminator enums.



Polymorphic Association Architecture & Security Perimeter

Engineering trade-offs for high-concurrency exam execution and candidate data protection.

1. Polymorphic Reference Architecture Decision (test_session_item & item_usage_log):

In the Item Bank data model, test_session_item.item_id and item_usage_log.item_id must point to four structurally heterogeneous item entities (PersonalityItem, DerailerItem, CognitiveItem, and SjtScenario). Two primary relational patterns were evaluated:

- **Approach A (Synthetic Discriminator + Unconstrained Foreign ID — RECOMMENDED):**

Store item_type (ItemType enum) and a single item_id (BIGINT) with composite indexes (test_session_id, item_type, item_id). Integrity is enforced by the Spring Boot ItemSelectionEngine.

- **Approach B (Sparse Nullable Foreign Keys):**

Maintain four separate nullable FK columns (personality_item_id, derailer_item_id, cognitive_item_id, sjt_scenario_id) with an SQL CHECK constraint ensuring exactly one FK is non-null.

Architectural Recommendation & Justification:

We strongly recommend **Approach A (Synthetic Discriminator Enum + Item ID)**. During live assessments with hundreds or thousands of simultaneous candidates submitting responses, session item queues and audit usage logs are write-heavy, high-throughput append structures. Decoupling session runtime tracking from rigid RDBMS multi-table foreign key cascades prevents table-level lock contention on the master content banks (personality_item, sjt_scenario). Furthermore, this approach maps directly to high-speed Redis session caching (where items are hydrated as lightweight polymorphic JSON payloads) without ORM join overhead. Domain-level referential integrity is validated at assembly time by the Spring Boot Item Selection Engine.

2. Zero-Knowledge Security Perimeter (Strictly Internal Fields):

To prevent inspection attacks, candidate cheating, and reverse-engineering of scoring algorithms, the following fields are declared **STRICTLY INTERNAL** and must NEVER be exposed across any candidate-facing API response:

- personality_item.is_reverse_keyed & ideal_target: Inverts Likert polarity and calibrates standard deviation.
- derailer_item.ideal_target: Identifies optimal vs. high-risk score cutoffs for dark traits.
- cognitive_item.correct_option_letter: The answer key (A/B/C/D) for GCAT cognitive questions.
- sjt_option.expert_rating & sjt_scenario_analysis.best_option_letter: Expert consensus benchmarks (1.00..4.00) used for Kendall-tau rank concordance calculation.

Implementation Rule: Candidate DTO mappers (MapStruct / dedicated CandidateItemResponse DTOs) must omit these fields at the controller perimeter.

3. Open Questions for Implementation Phase:

- **Q1 — SJT Response Payload Structure:** The table list defines sjt_option.expert_rating (DECIMAL) and sjt_scenario_analysis.best_option_letter. In the scoring engine specification, SJT uses a 4-option full ranking format (Kendall-tau rank correlation τ). We confirm that candidates submit a 4-element ordered permutation (e.g. ['C', 'A', 'B', 'D']) rather than a single choice.
- **Q2 — Item Exposure Counter Concurrency:** exposure_count and last_used_at on content items will be updated asynchronously via Redis buffer or batch job to prevent database row locks during active exam sessions.
- **Q3 — Cognitive Media Attachments:** Abstract cognitive items often require vector diagram SVGs or image assets. We recommend adding an optional media_asset_url column on cognitive_item during Phase 1 JPA entity creation.